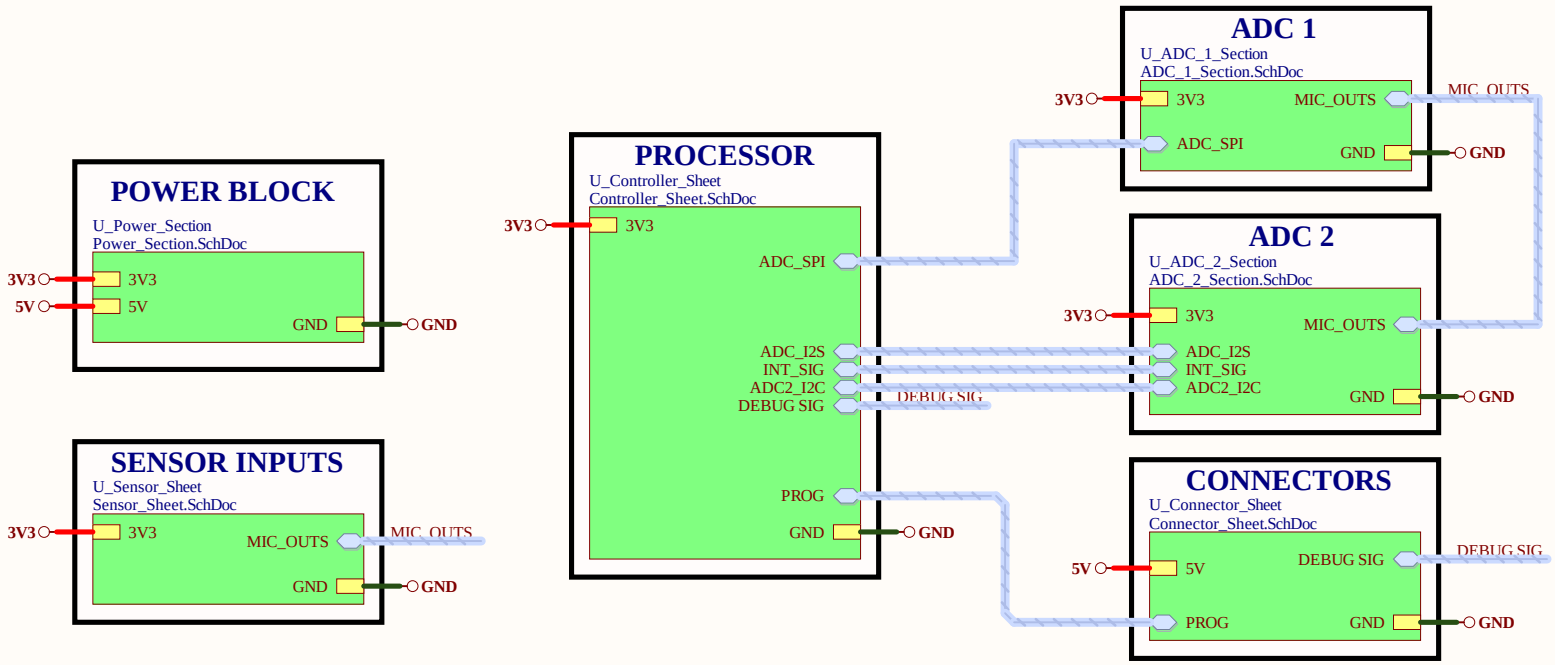
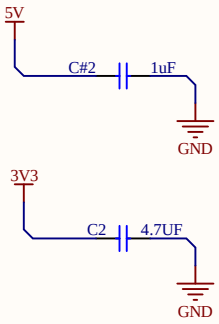
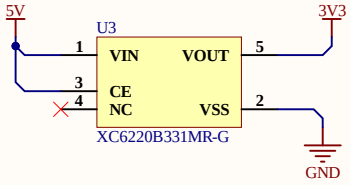


Cannot open file C:\Users\user\Pictures\Saved Pictures\espark-consulting-group-150x150.jpg. File does not exist.	Title: Sound_Sensor.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 12/14/2022	Sheet 1 of 7	
	File: C:\Users\..\Top_Sheet.SchDoc	Drawn By: Huzaifa Ali	

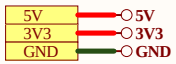
DECOUPLING CAPS



3.3V LDO



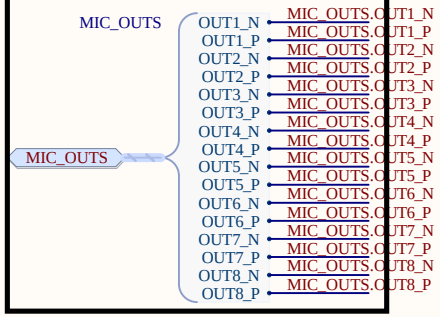
Fixed 3.3V output
150mV dropout @500mA
Max output current is 1A
Enable High to activitate the chip



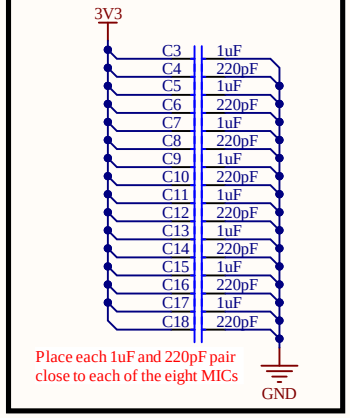
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D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Power_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/14/2022	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 7

MIC OUT SIGNALS

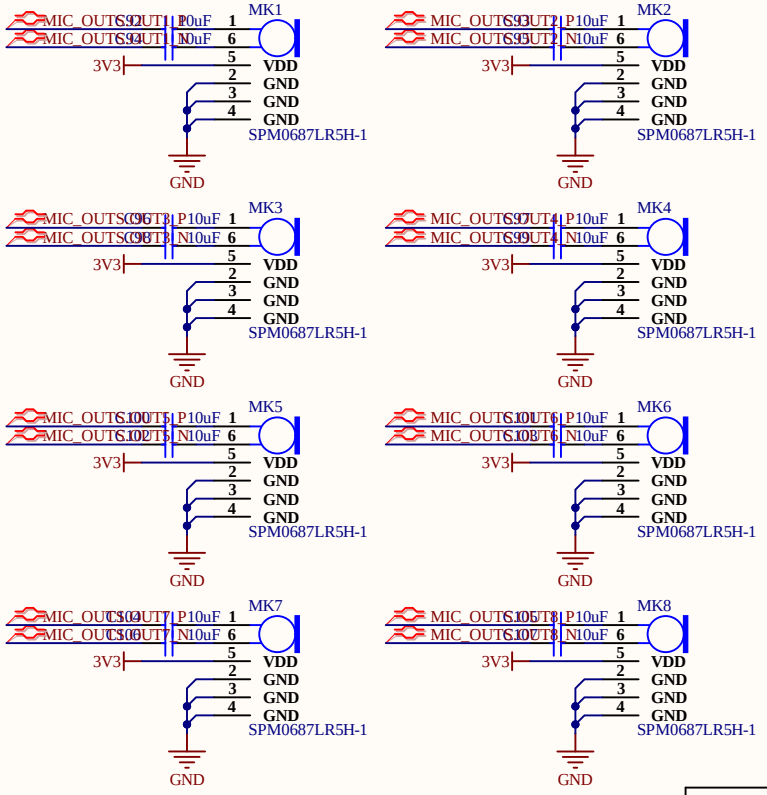


DECOUPLING CAPS





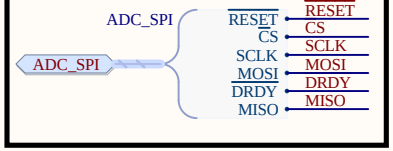
ANALOG MIC INPUTS



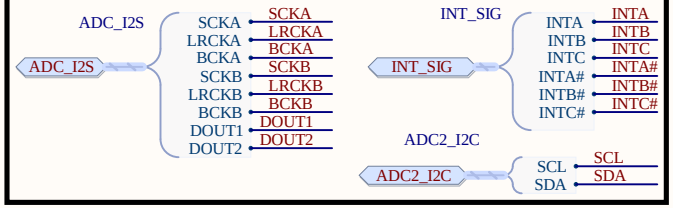
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Sensor_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/14/2022	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 7

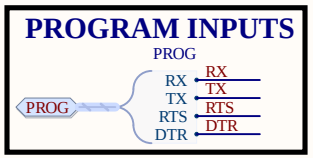
ADC1 OUTPUT



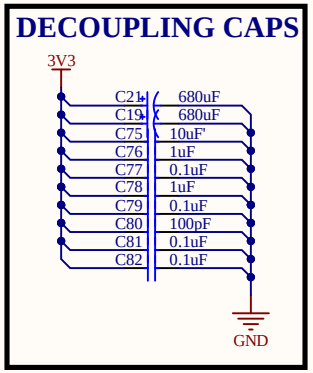
ADC 2 OUTPUT SIGNALS



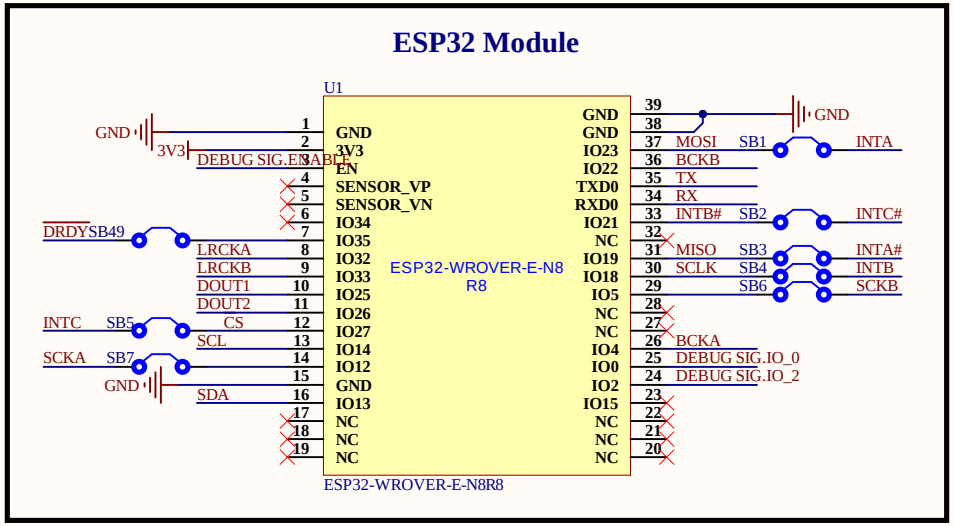
PROGRAM INPUTS



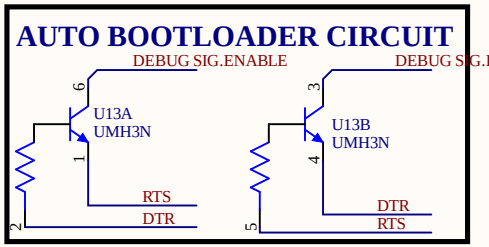
DECOUPLING CAPS



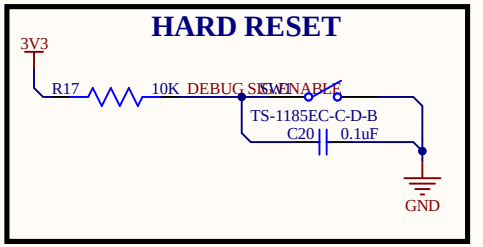
ESP32 Module



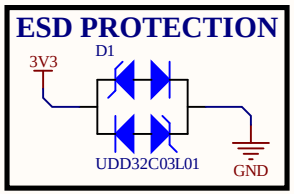
AUTO BOOTLOADER CIRCUIT

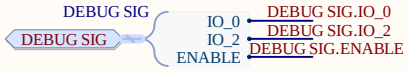


HARD RESET



ESD PROTECTION



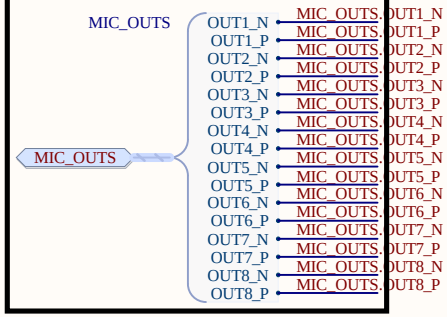




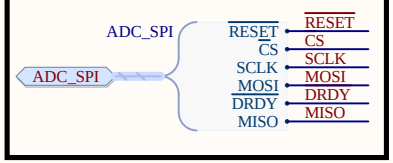
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D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Controller_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/14/2022	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 7

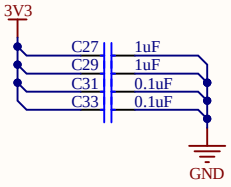
MIC OUT SIGNALS



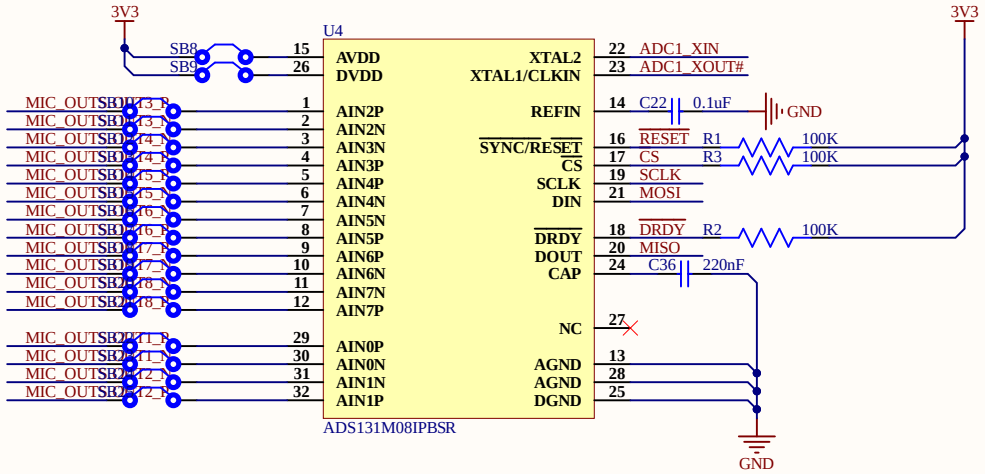
ADC OUTPUT



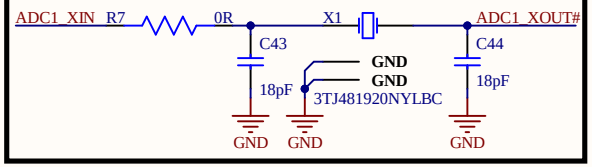
DECOUPLING CAPS



ADC WITH SPI INTERFACE



8.192 MHz Crystal





Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: ADC_1_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/14/2022	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 7

I2C CONTROL SIGNALS

ADC2_I2C

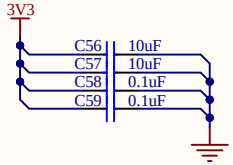
ADC2_I2C — SCL — SCL; SDA — SDA

MIC OUT SIGNALS

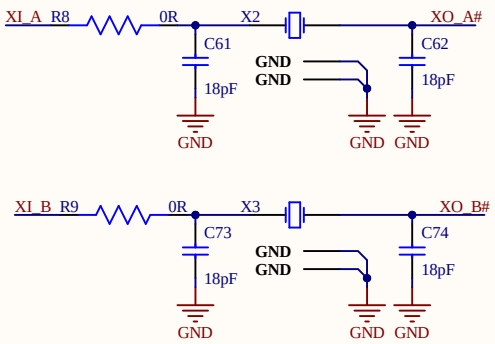
MIC_OUTS

MIC_OUTS — OUT1_N … OUT8_P — MIC_OUTS.OUT1_N … MIC_OUTS.OUT8_P

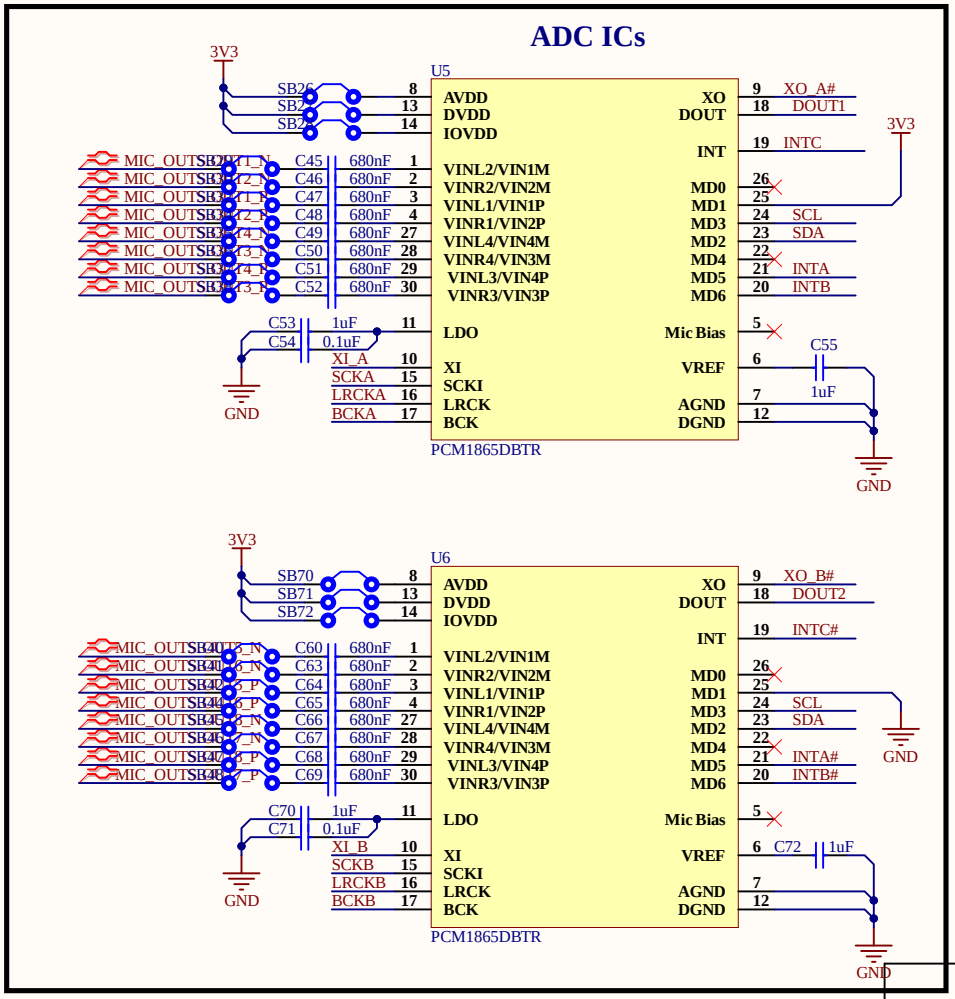
DECOUPLING CAPS



24.576 MHz Crystals



ADC ICs



ADC_I2S — SCKA LRCKA BCKA SCKB LRCKB BCKB DOUT1 DOUT2

INT_SIG — INTA INTB INTC INTA# INTB# INTC#

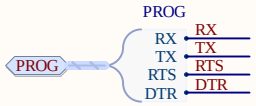
3V3 — 3V3

GND — GND

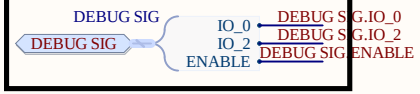
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Title: ADC_2_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/14/2022	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 7

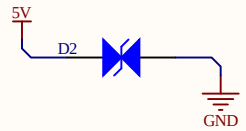
PROGRAM INPUTS



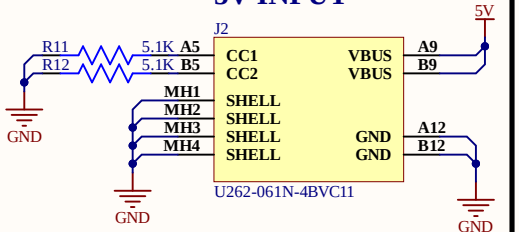
DEBUGGING SIGNALS



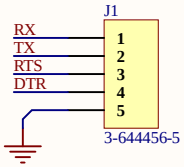
ESD PROTECTION



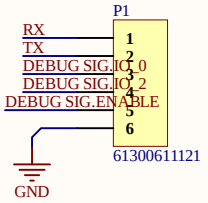
5V INPUT



PROGRAMMING HEADER



DEBUG CONNECTOR





Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Connector_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/14/2022	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 7